



**Generation Interconnection
System Impact Study Report
for**

Queue Project AE1-225

MILTON-SUNBURY 69KV

9.4 MW Capacity / 19.8 MW Energy

August, 2019

1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

2 General

The Interconnection Customer (IC) has proposed a Solar generating facility located in Snyder County, Pennsylvania. The installed facilities will have a total capability of 19.8 MW with 9.4 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is September 30, 2020. **This study does not imply a TO commitment to this in-service date.**

Queue Number	AE1-225
Project Name	MILTON-SUNBURY 69KV
Interconnection Customer	Talen Generation LLC
State	Pennsylvania
County	Snyder
Transmission Owner	PPL
MFO	19.8
MWE	19.8
MWC	9.4
Fuel	Solar
Basecase Study Year	2022

2.1 Point of Interconnection

AE1-225 will interconnect with the PPL EU transmission system via the Sunbury – Milton 69 kV line, 1.35 miles from the Sunbury bus and 9.11 miles from the Milton bus. The POI will be where the PPL EU tap line terminates (with insulators) at the first dead-end structure inside the IC substation.

The scope of work provided in this Feasibility Study is for facilities to be constructed on the PPL EU side of the POI.

2.2 Cost Summary

The AE1-225 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 1,115,000
Direct Connection Network Upgrade	\$ 0
Non Direct Connection Network Upgrades	\$ 230,000
Total Costs	\$ 1,345,000

In addition, the AE1-225 project may be responsible for a contribution to the following costs:

Description	Total Cost
System Upgrades	\$ 0

3 Transmission Owner Scope of Work

3.1 Attachment Facilities

69kV Transmission Line Tap

The Attachment Facilities will interconnect with the Milton-Sunbury 69 kV line. This scope of work is based on the IC collector substation GPS Coordinates: 40°51'17.43"N, 76°50'18.65"W.

- Tap the existing Sunbury-Milton 69 kV (built for 230 kV) line.
- Replace existing tower 25229S55557 with new tap two pole tap structure.
- Extend tap to customer substation dead-end structure. Design tap for initial and final single circuit.
- Install new MOLBAB on customer tap. The pole to support the Montour-Sunbury 230kV line will not be designed to be tapped. Install 556 ACSR conductor on the tap and single OPGW.

Transmission Fiber

- Install splice box on new tap pole to AE1-225 at 25229S55557.
- Complete a fiber throw on the parallel Sunbury – Columbia 69kV line between structures 25239S55554 and 25241S55612. Install new splice box at 41/12. Install new splice box at 25239S55554
- Run ADSS to the splice box on the new tap pole at 29/57.

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Total Attachment Facility Costs	\$ 1,115,000

3.2 Direct Connection Cost Estimate

None

3.3 Non-Direct Connection Cost Estimate

Remote End work – Sunbury and Milton Substations

- Model IC in CAPE and conduct a wide area short-circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed.
- Conduct a review of the IC relay settings and engineering package (submitted by IC to PPL EU).
- The Sunbury-Milton 69 kV line is operated in network and thus the scope of work is required at both remote end substations
- The following upgrades are required at **Milton** Substation Remote End
 - Install new fiber-based DTT equipment
 - Modify the existing Sunbury 69kV #14L circuit breaker protection and control scheme.
 - Modify the existing SCADA for new alarms
 - Modify the existing Alarm Management System (AMS).
 - Install new cables and modify control wiring for the above.
 - Perform system checks and test equipment before placing in service.
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- The following upgrades are required at the **Sunbury** Substation Remote End:
 - Install new fiber-based DTT equipment
 - Modify the existing Milton 69kV #2S circuit breaker protection and control scheme.
 - Modify the existing Fairview Tie 69kV #2T circuit breaker protection and control scheme.
 - Modify the existing SCADA for new alarms.
 - Modify the existing Alarm Management System (AMS).
 - Install new cables and modify control wiring for the above.
 - Perform system checks and test equipment before placing in service.

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Remote End Work – Milton Substation	\$ 115,000
Remote End Work – Sunbury Substation	\$ 115,000
Total Non-Direct Connection Facility Costs	\$ 230,000

4 Schedule

The estimated time to complete the scope of work is **12 - 18 months** after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

Assumptions

- Availability of optimal transmission line route
- Outage feasibility not assessed until Facilities Study
- No major environmental, real estate, or permitting issues
- IC is responsible for acquisition of easements and right of way for the Attachment Facilities

5 Interconnection Customer Requirements

5.1 PPL EU IC Direct Transfer Trip (DTT) Requirements

PPL EU will require an independent communication path, for DTT of the IC Intertie Protective Relaying (IPR) Fault Interrupting Devices (FIDs), consisting of one communication circuit with both the Sunbury 230/69 kV substation and the Milton 230/69 kV substation. To ensure reliable communication, the IC shall also provide DTT relaying equipment identical to the PPL EU DTT relaying equipment. All DTT relaying equipment shall connect to the respective communication path. All DTT relaying equipment should reside within the same location as the IPR and Point of Contact (POC) relaying equipment.

PPL EU currently has OPGW on the Sunbury-Columbia 69 kV line adjacent to the requested interconnection Sunbury-Milton 69 kV line available for DTT to the Sunbury and Milton substations. PPL EU assumes strands of this fiber will be used for the independent communication pathway. However, the IC may choose to procure a third-party communication circuit at its own discretion in lieu of the OPGW.

5.2 PPL EU Interconnection Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

6 Revenue Metering and SCADA Requirements

6.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

6.2 PPL Requirements

Installation of revenue grade Bi-directional Metering Equipment will be required in the vicinity of the POI to measure kWh and kVARh. PPL EU will design and supply the required metering equipment; all installation costs would be borne by the IC including CTs/PTs. All metering equipment must meet applicable PPL EU tariff requirements as well as being compliant with all applicable requirements of the PJM agreements. The equipment must provide bidirectional revenue metering (kWh and kVARh) and real-time data (kW, kVAR, circuit breaker status, and generator bus voltages) for the IC's generating resource. The metering equipment should be housed in a control cabinet or similar enclosure and must be accessible to PPL EU metering personnel.

7 Network Impacts

The Queue Project AE1-225 was evaluated as a 19.8 MW (Capacity 9.4 MW) injection at **Milton-Sunbury 69kV line** in the PPL area. Project AE1-225 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-225 was studied with a commercial probability of 1.00. Potential network impacts were as follows:

Summer Peak Load Flow

8 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

9 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

10 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

11 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

12 System Reinforcements

None

13 Stability and Reactive Power Requirement for Low Voltage Ride Through

Stability Analysis is not required for AE1-225.

14 Light Load Analysis

No mitigations required.

Short Circuit

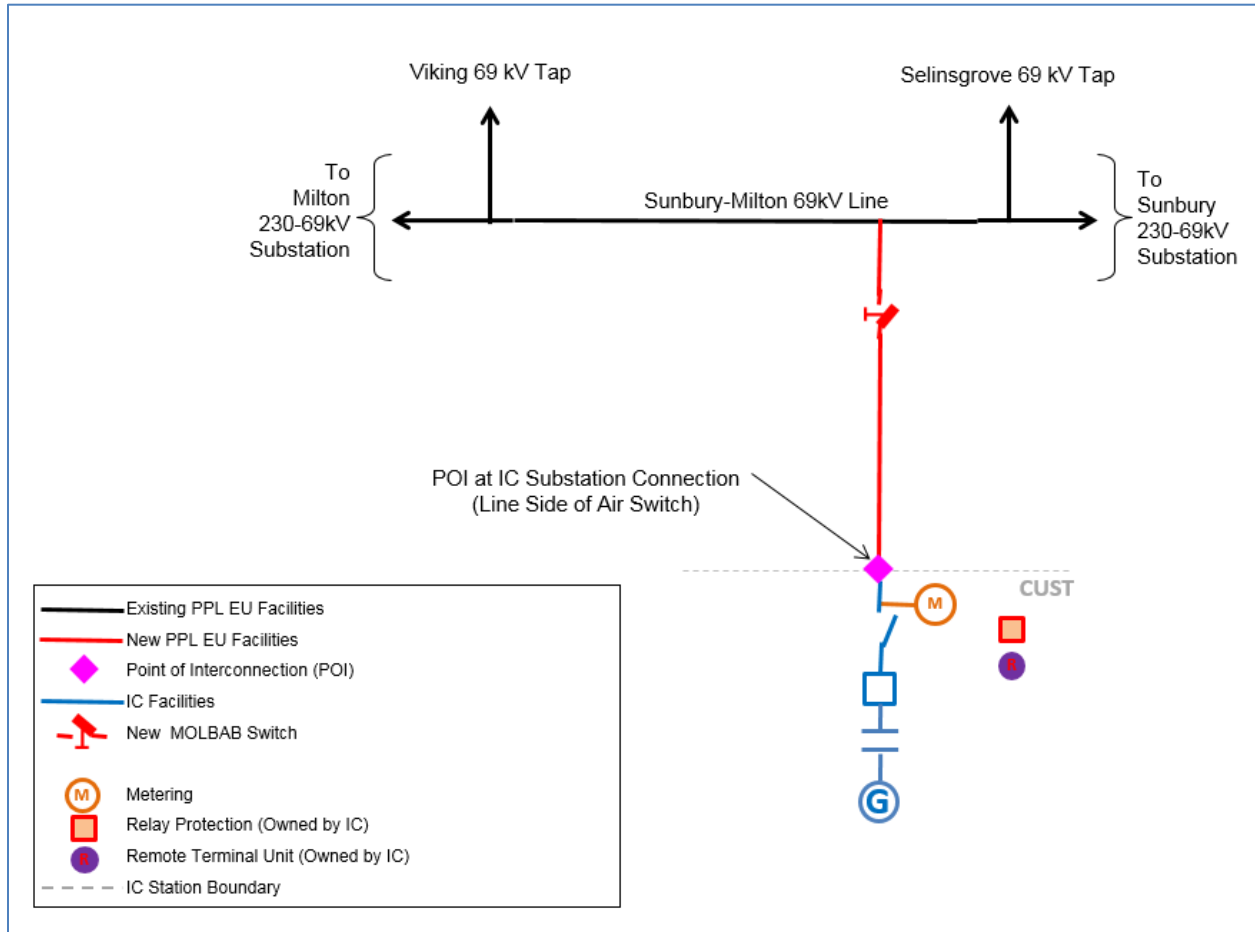
15 Short Circuit

The following Breakers are over duty

None

Attachment 1

Single Line Diagram



Attachment 2

Project Location

